

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to transform how conceptual understanding is evaluated by leveraging voice input and NLP techniques. The system will allow users to upload or record audio explanations of a given concept, which will be transcribed, analyzed, and compared with reference content to produce detailed feedback and a similarity score.

Project Overview	
Objective	The primary objective is to build an AI-powered system that evaluates user explanations of concepts through voice, understands the content, and provides accurate scoring, feedback, and improvement suggestions.
Scope	The project covers audio upload/recording, speech-to-text conversion, NLP-based semantic analysis, concept matching, scoring, feedback generation, and reporting via a web application.
Problem Statement	
Description	Students and educators lack efficient tools to evaluate conceptual understanding from spoken explanations. Manual evaluation is time-consuming, subjective, and does not provide consistent, detailed insights.
Impact	Existing evaluation methods are slow, provide generic feedback, and do not help learners identify specific gaps in understanding. This reduces learning effectiveness and motivation.
Proposed Solution	
Approach	Use AI models for speech recognition, semantic understanding, and similarity scoring to evaluate explanations. Provide detailed feedback and improvement tips through an intuitive web application.
Key Features	- Voice-based explanation input (upload/record) - Accurate transcription using Speech-to-Text - NLP-based semantic analysis and concept matching - Detailed scoring and concept-wise feedback - Progress tracking and report generation - Real-time feedback with improvement suggestions - Comparison of explanation with ideal reference answers
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i7 / Ryzen 7, 8 cores NVIDIA RTX 3060 (6GB VRAM)
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	FastAPI, scikit-learn, PyTorch
Libraries	Additional libraries	Transformers, Whisper, spaCy, NLTK, pandas, numpy, matplotlib
Development Environment	IDE	VS Code, Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle datasets, ConceptNet, custom concept datasets (~10 GB, CSV/JSON/MP3)